

AMMAR HASSAN

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EDUCATION

COMSATS University, Lahore

Bachelor of Science in Software Engineering

2021 - 2026

Lahore, Pakistan

International School Lahore, DHA Campus

A-Levels (Advanced Level)

2019 - 2021

Lahore, Pakistan

EXPERIENCE

Corporate Events Company (Client)

Freelance Web Developer

Jul 2026 – Present

Lahore, Pakistan

- Rebuilding their **multi-page marketing site** end to end, working directly with the client through requirements gathering, design review rounds and revisions.
- Built with **Astro** and deployed on **Cloudflare Pages**, with a shared motion system, SEO-targeted service pages and layouts verified at desktop and mobile breakpoints.

Punjab Safe Cities Authority (PSCA) 🔗

Web Development Intern

Jun 2024 – Aug 2024

Lahore, Pakistan

- Supported the maintenance of **citizen-facing government portals**, gaining practical experience in optimizing **Node.js** architectures for high-traffic provincial services.
- Gained architectural insight into the system design of Punjab's **Automated E-Challan Network**, tracing the data pipeline from AI-based camera detection to web-dashboard visualization.

PROJECTS

CUI Central - Campus Ecosystem 🔗 | *Next.js, Socket.IO, Gemini AI, MongoDB*

- Architected a multi-tenant campus ERP consolidating **7+ core services** (Digital Cafeteria, Job Portal, Room Booking) into a unified **Next.js 15** monorepo. Implemented complex **RBAC** (Role-Based Access Control) for secure Admin, Faculty, and Student workflows.
- Engineered a real-time infrastructure using **Socket.IO** for live faculty availability tracking and order status updates. Integrated a **RAG-based AI Agent** (Google Gemini) with MongoDB Vector Search to automate student support.
- [Live Site](#)

Cloud Visitor Counter - Serverless Backend 🔗 | *AWS, Terraform, Python, GitHub Actions*

- Architected a serverless backend tracking live site telemetry on AWS (Lambda, API Gateway, DynamoDB), provisioning all infrastructure via **Terraform** (IaC).
- Engineered a secure CI/CD deployment pipeline using **GitHub Actions** and passwordless **OIDC** authentication, ensuring atomic increments and safe remote state management.
- [GitHub](#)

Check! - Multiplayer Game 🔗 | *Next.js, Node.js, Socket.IO, XState*

- Engineered a custom WebSocket game engine supporting concurrent rooms. Achieved **<50ms latency** for game state synchronization using event-driven **Socket.IO**. Implemented **XState** state machines to manage complex game logic and solve race conditions, ensuring game integrity across multiple clients.
- [Live Site](#) | [GitHub](#)

Karting Analysis - Telemetry Pipeline 🔗 | *Python, Pandas, NumPy, Matplotlib*

- Built a Python pipeline to ingest and clean **50,000+ telemetry points**. Optimized data processing speed by **30%** by replacing iterative loops with **NumPy** vectorization. Generated statistical heatmaps to visualize driver consistency and degradation.
- [Live Site](#) | [GitHub](#)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL

Web & Frameworks: Next.js, React, Node.js, Express, Socket.IO, Tailwind CSS

Data & AI: Pandas, NumPy, Matplotlib, RAG, Vector Search, Gemini API

Cloud & Tools: AWS, Terraform, Docker, Git, GitHub Actions, MongoDB, PostgreSQL, Linux